

# Doctoral Position

## Modelling and Control of Near-Wall Turbulence: From Physical Understanding to Machine Learning Approaches

Institut Pprime, CNRS, Poitiers, France

Supervisors: Dr. Lionel Agostini & Dr. Laurent Cordier

### Position Details

- 📅 **Duration:** 36 months
- 📍 **Location:** Institut Pprime, CNRS, Poitiers, France
- 💶 **Salary:** €2,000 net/month
- 👤 **Scientific Supervisors:** Dr. Lionel Agostini & Dr. Laurent Cordier
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Nowadays, experimental and numerical simulations progressively provide an unprecedented **volume of extremely detailed data**, which need to be examined and interpreted. There is therefore an increasing urgency of refined investigative tools for appropriate statistical analyses and data mining. **Machine-learning (ML) algorithms** offer a new path for investigating high-dimensional, nonlinear problems, such as near-wall Turbulence. The development of ML methods, associated with the abundance of data and combined with **solid background in Turbulence**, offer a unique opportunity for achieving major breakthrough in terms of advances in wall-bounded flows and their control. The key objectives of this research programme, by using a large spectrum of statistical analysis combined with various ML approaches, is to nail the **boundary-layer theory** down and derive **models** that are able to predict relevant physical characteristics and the response to **forcing near-wall Turbulence** for both incompressible and compressible flows, where density variations are caused by pressure or temperature fluctuations or both. The other objective is to provide solutions for **reducing drag** and, in the case of a thermal boundary layer, to **maximise the heat transfer**, whilst keeping losses as low as possible.

### Keywords

Wall-bounded flow, Turbulence, Drag, Heat transfer, Control, Machine learning, Auto-encoder, Reduced order model

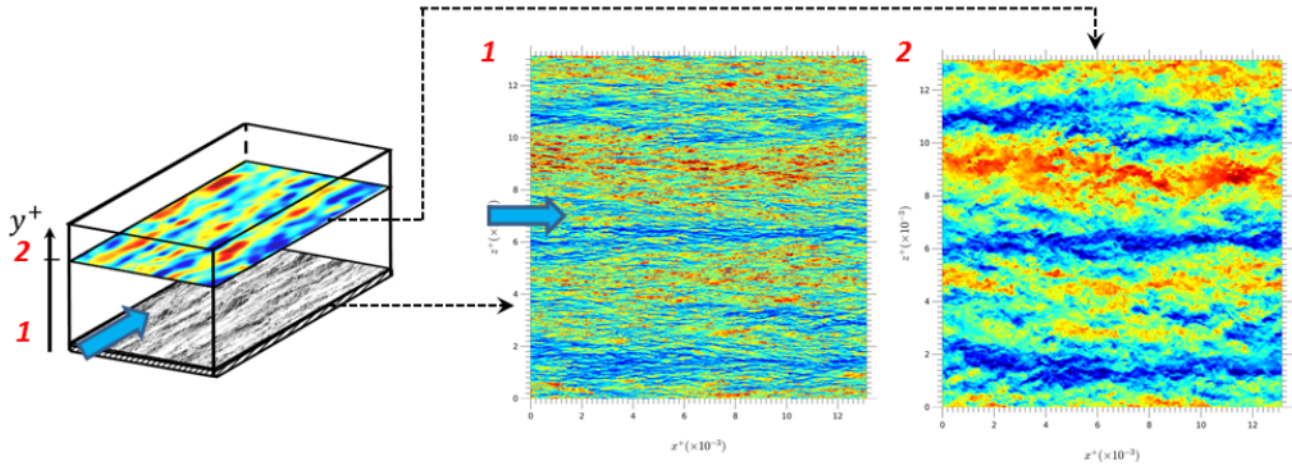


Figure 1: Streamwise velocity field for  $Re_\tau \approx 4200$ . 1: in the viscous sublayer and 2: in the outer flow (mid-height of the log law).

## Societal Context

Turbulent flows dictate the performance characteristics of numerous industrial equipment and environmental applications. One important consequence of Turbulence is high friction drag on surfaces, the increase relative to laminar conditions easily reaching factors of 10-100, depending on the Reynolds number of the flow. In many applications, the friction drag is extremely influential to the operational effectiveness of the device or process. This applies especially to transport, involving either self-propelling bodies moving in a fluid or fluids being transported in ducts and pipes. The United Nations' International Civil Aviation Organization (ICAO) expects aviation emissions to roughly triple by 2050, at which time aircraft might account for 25% of the global carbon budget (ICCT). There is, therefore, significant pressure to **reduce transport-related emissions, of which friction drag is a major constituent**.

On the other hand, enhancing the turbulent fluxes within the wall-bounded region, is generally beneficial for the heat transfer at walls. Thus, in the case of heat exchangers, a balance need to be found between drag-induced losses and the heat transfer. With the development of renewable energy, increasing the efficiency of heat exchangers is a fast-growing field of research, especially for solar receivers.

## Scientific Challenges

In contrast with many scientific fields (brain functioning for instance), fluid mechanics is in the enviable position of having a first principle equation, the Navier-Stokes (NS) equations, to describe flows. In principle, the flow motions can be perfectly predicted by solving these equations (Laplace's determinism). However, the NS equations prove to be extremely complex and even impossible to resolve in some cases, mainly because of its non-linearity.

For any viscous fluid in motion relative to a solid, the velocity decreases to zero at the wall inducing a shear layer. The flow produced by this shear layer creates chaotic turbulent structures within a wide spectrum of length and time scales. The stronger the shear layer is,

richer and more complex is the dynamics, which renders the study of near-wall Turbulence extremely fascinating and challenging.

Since Prandtl at the beginning of the last century, near-wall flows have intensively been studied, however more than two centuries after equations governing turbulent flows were introduced, our **understanding of near-wall Turbulence remains pitifully modest**. This is all more true since as the Reynolds numbers increases turbulent flow becomes more and more complex as new families of coherent structures are produced and enrich the flow dynamics. Without a sufficiently detailed understanding of turbulent flows, most control strategies remain sub-optimal.

Thanks to recent advances in metrology, and the drastic reduction in computational costs, both the number and the size of database have undergone substantial growth in fluid mechanics. This abundance of data poses new challenges for the postprocessing that requires appropriate tools. Thankfully, many techniques have been developed by the **Machine Learning (ML) community** that are readily applicable to spatiotemporal fluid data, and are sustained by the recent advances in hardware (development of GPU/TPU platforms), algorithms and open-source libraries.

By definition, ML is the field of study that gives computers the ability to learn without being explicitly programmed. In this project, Machine Learning algorithms will not aim to replace knowledge from physics (evolution laws for instance) with data, but rather to **use models and data in synergy**. Indeed, it would be completely nonsense to get rid of centuries of scientific knowledge or proven models. The applicant will therefore use Machine Learning methods as complements to traditional approaches by imposing, for example, symmetry properties or known evolutionary models.

**The identification of the phenomena driving the flows dynamics remains one of the major challenges of physics and engineering today**. This complexity has traditionally been addressed by statistical Turbulence models, but these are proven to have only limited generality, even in the case models are based on the most elaborate transport equations for the second moments of the turbulent fluctuations. The recent developments of **Machine Learning** methods for the **modelling of large-scale problems and their control** prove to be a promising way forward.

Based on our current knowledge of turbulent flows, and from the many progress which still to be made using Machine Learning techniques, one can expect to introduce more effective new control methods. Moreover, any efforts towards unravelling the nature of structures populating boundary layers, and towards understanding their dynamics, are crucial to progress towards **universal modelling of Turbulence**, and contribute for deriving effective **control systems** that are compatible with the requirements of industrial applications.

## Objectives

Considerable progress in understanding near-wall flows and their control can be accomplished by capitalising on a solid foundation in Turbulence, augmented by the novel strategies introduced by Machine Learning. The main ambition is to generate a **low-dimensional dynamical model** pertaining to near-wall turbulence. In the initial phase,

an auto-encoder will be utilised to identify a more **compact coordinate system** for the dynamical system. One of the first objectives is to construct this auto-encoder and comprehend its functionality, -i.e be able to interpret it. Thus, the primary aim is to lay down a framework or “skeleton” for the low-dimensional dynamical model. The subsequent goal involves discerning how the disparate segments of this “skeleton” are interconnected, to shed the light on the **dynamical** aspect. To fulfil this, an array of machine learning techniques, including clustering and transformers, will be employed.

## Your Profile

-  Master's degree in Fluid Mechanics, Applied Mathematics, Physics, or related field
-  background in computational methods and programming
-  Knowledge of fluid mechanics and interest in turbulence physics
-  Excellent communication skills and ability to work in a collaborative environment
- Motivation to work at the intersection of fundamental research and applied engineering

## About the Laboratory

The Pprime Institute is a CNRS Research Unit featuring interdisciplinary expertise across fluid mechanics, materials science, and mechanical engineering. You will join the Curiosity team, which focuses on innovative approaches to fluid mechanics challenges using advanced computational and data-driven methods.

**Note:** The laboratory is classified as a Zone à Régime Restrictif (ZRR), meaning that recruitment is subject to prior authorisation by the Defence Security Officer.

## How to Apply

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For more information and to submit your application, please send an email to:

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